CS7.503: Mobile Robotics — Mid Submission

Lifelong Mapping & 3D Change Detection

Ch Shreyash 2024121008 | Farzana S 2024121006 | Prajas Wadekar 2024121009

Code: https://github.com/PrajasW/LM-3DCD-SG

Abstract

A robot that returns to a room with a map from an earlier visit has to find its pose in that map with no initial guess, decide which objects stayed, moved, appeared or disappeared, and edit the map accordingly. We pose this as joint relocalization and change detection on object-level 3D scene graphs, and propose a training-free pipeline built on one observation: objects that did not move keep their mutual distances, so the largest mutually consistent set of object matches is the static set and gives the pose, while smaller consistent sets are objects that moved together. This report covers Phase 1: scene graphs from 3RScan/3DSSG annotations, Steps 1–3 (candidate matches, maximum consistent set, pose), and a two-visit evaluation on the full 3RScan validation split (47 rooms, 110 pairs) in which the new visit is moved by an unknown rigid transform. Our method relocalizes 89.1% of the pairs at the RIO threshold (20 cm, 20°) and returns the fewest wrong poses of all methods (5 of 110), in 0.25 s per pair including graph construction. Dense FPFH+RANSAC reaches 93.6% at about 20× our runtime; the difference is not significant on 110 pairs. The learned aligners SGAligner and SG-PGM drop to 32% and 34% under the transform (from 79% and 75% on pre-aligned frames).

1 Problem Statement

Motivation. Robots in homes, offices and labs return to the same rooms many times, and the rooms change in between. On return, the robot must relocalize in its old map although some landmarks have moved, tell moved, added and removed objects apart from objects that are only hidden or out of view, and update the map. Replacing the old map with the new scan deletes everything that is not currently visible, loses object identities and moves the map frame; appending every scan keeps outdated copies and makes the map grow with every visit. Object-level 3D scene graphs [20, 29] allow per-object edits, but, as Section 2 shows, no existing method relocalizes without an initial guess, labels per-object changes and edits a persistent map without training.

Map and observations. The map is an object-level scene graph $\mathcal{G}^M = (\mathcal{V}^M, \mathcal{E}^M)$ in a fixed map frame. Each object node $o \in \mathcal{V}^M$ stores its points $P_o \subset \mathbb{R}^3$, centroid $\mathbf{c}_o$, extents $\mathbf{e}_o \in \mathbb{R}^3$, class ℓ_o, height above the floor h_o and change history; structural nodes hold the floor and walls; edges carry support and spatial relations. A visit produces a point cloud split into segments, from which we build $\mathcal{G}^N = (\mathcal{V}^N, \mathcal{E}^N)$ in the robot's own frame, related to the map frame by an unknown $\mathbf{T}^\star \in \mathrm{SE}(3)$. There is no initial guess and no odometry between visits.

Problem. Given $\mathcal{G}^M$, $\mathcal{G}^N$ and the visit's depth images, estimate

(P1) the pose $\hat{\mathbf{T}} \in \mathrm{SE}(3)$ such that $\hat{\mathbf{T}}\,\mathbf{c}_{o'} \approx \mathbf{c}_o$ for every unchanged pair (o, o'), or report failure ($\perp$) when the observation does not determine it;

(P2) a partition of the objects into *static* matches $S \subset \mathcal{V}^M \times \mathcal{V}^N$ (one-to-one), *moved* groups $\{(M_k, \mathbf{T}_k)\}$ with their rigid motions, *removed* $R \subset \mathcal{V}^M$, *added* $A \subset \mathcal{V}^N$ and *not observed* $U \subset \mathcal{V}^M$, where R and A require depth evidence of free space and every other unexplained object goes to U;

(P3) the updated map $\mathcal{G}^{M'}$, obtained by editing $\mathcal{G}^{M}$ per object so that identities persist and $|\mathcal{G}^{M'}|$ grows with the number of objects, not the number of visits.

Over visits $t = 1, \ldots, K$ the map is the result of applying this update repeatedly.

Phase-1 sub-problem and metrics. With segments from the 3RScan/3DSSG annotations, solve (P1) and the static part of (P2): return $(\hat{\mathbf{T}}, S)$ or $\perp$. A pair is relocalized if the rotation error $\angle(\hat{\mathbf{R}}, \mathbf{R}^\star) \leq \tau_r$ and the translation error $\|\hat{\mathbf{t}} - \mathbf{t}^\star\| \leq \tau_t$, with the RIO thresholds $(\tau_t, \tau_r) \in \{(10\,\text{cm}, 10^\circ), (20\,\text{cm}, 20^\circ)\}$ [28]; $\perp$ counts as a failure. We also report registration recall (RMSE of the visit's points $< 0.2\,\text{m}$, the criterion of [22, 31, 13]) and runtime.

Refinements over the proposal. (i) Declining ($\perp$) is part of the output and counted as a failure, so a method cannot gain by guessing. (ii) Descriptors must be pose invariant by construction (Section 3.2). (iii) Registration recall is added so that our numbers are comparable with the scene-graph alignment literature. (iv) Every method, learned or not, sees the same randomly transformed visits.

2 Literature Review

3D scene graphs. Armeni et al. [2] introduced layered 3D scene graphs. Kimera [20] and Hydra [10] build them online from SLAM, with objects assumed static and only agents moving. 3DSSG [29] annotates every 3RScan scan with an object-level graph (534 classes, 41 relations) and learns to predict it; SceneGraphFusion [30] predicts graphs incrementally from RGB-D, and ConceptGraphs [9] attaches open-vocabulary features to object nodes. Their update rules either assume a static world or delete objects that are not re-detected, whether or not they were in view. We reuse the representation but make the per-visit edit explicit and evidence based.

Scene-graph alignment and relocalization. RIO [28] introduced 3RScan and object instance relocalization across rescans; X-View [8] localizes with semantic graphs built from views. SGAligner [22] learns multi-modal node embeddings (points, relations, attributes), matches the top-K nodes and registers the matched objects with GeoTransformer [19] and GC-RANSAC [3]. SG-PGM [31] casts alignment as partial graph matching, fuses point-level features into the nodes, and re-scores point correspondences with node matches. SG-Reg [13] targets generalization and efficiency with a lighter network and a robust back-end; SGAligner++ [26] adds language features, and OpenSGA [5] moves to open-world graphs. All of these are trained, are evaluated on sub-scans cut from the same scan, and ignore unmatched objects, so they do not output changes. ROMAN [18] aligns open-set object maps for global localization with CLIPPER [15]; it is the closest training-free relative of our Step 2.

Consistency-based data association and robust registration. Joint compatibility [17] and PCM [16] select the largest set of mutually consistent measurements (PCM through a maximum clique, for loop closures between maps); CLIPPER [15] relaxes this to a weighted dense subgraph; TEASER [32] and maximal-clique registration [33] apply the pairwise-invariant test to point correspondences. Classical global registration pairs FPFH features [21] with RANSAC [7] and refines with ICP [4, 6]. We lift the pairwise test from points to objects, which shrinks the problem from thousands of correspondences to tens and makes the consistent sets meaningful: the static set versus objects moved together.

Change detection and long-term mapping. Objects Can Move [1] detects moved, added and removed parts between two *aligned* scans by geometric-transformation consistency and outputs unlabelled clusters. Living Scenes [34] relocalizes and reconstructs objects across 3RScan rescans, again given the scan alignment. DSG [12] labels objects stable, appeared or missing with Gaussian renderings and a language model, uses ground-truth poses and has no "moved" state; 3D VSG [14] predicts how likely an object is to change but does not verify it. Panoptic Multi-TSDFs [24] and Khronos [23] keep object-level volumetric maps consistent in changing scenes within one SLAM session, and LT-mapper [11] does LiDAR lifelong mapping with change detection after alignment.

Table 1: Capabilities of related systems (✓ yes, – no, ○ partial). "Pose" = relocalization without an initial alignment; "Changes" = per-object moved/added/removed; "Multi" = evaluated over > 2 visits.

Method	Pose	Changes	Map edit	Training-free	Multi
Kimera / Hydra [20, 10], ConceptGraphs [9]	–	–	○	✓	–
SGAligner [22], SG-PGM [31], SG-Reg [13]	✓	–	–	–	–
ROMAN [18]	✓	–	–	✓	–
Objects Can Move [1] / Living Scenes [34]	–	○	–	✓ / –	–
DSG [12]	–	○ (no moved)	✓	○ (LLM)	–
3D VSG [14]	–	predicted only	–	–	–
Ours (proposed)	✓	✓	✓	✓	✓

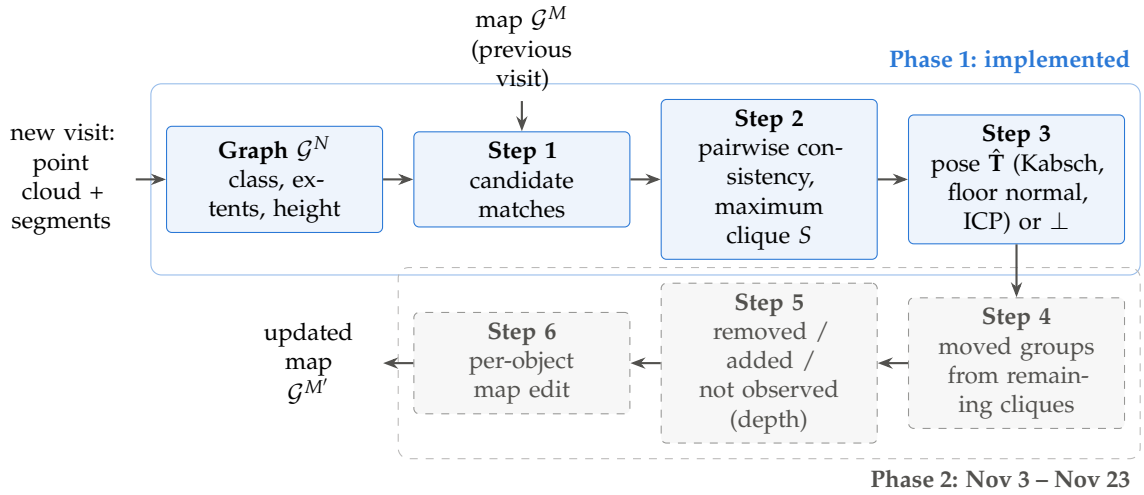


Figure 1: Pipeline for one visit. Phase 1 (blue) builds the visit's graph and relocalizes it in the map; the static set and the remaining consistent groups it produces are the inputs of Steps 4–6 (dashed, Phase 2).

Viability and motivation. Table 1 summarizes the gap: no method jointly relocalizes without an initial guess, labels per-object changes with free-space evidence, and edits a persistent map over many visits without training. The pieces are known to work separately: maximum-clique selection is robust to over 95% outliers at the point level [32, 33], 3RScan provides instance IDs shared across rescans together with the moved/removed annotations needed for evaluation [28], and Section 3.3 shows that the object-level version is already within the uncertainty of dense FPFH registration at a twentieth of its cost.

3 Progress So Far

Summary. All Phase-1 deliverables of the proposal are complete:

- **Data and graphs:** loaders for 3RScan (instance PLY, labels, rescan transforms, moved/removed annotations) and 3DSSG; object-level scene graphs with pose-invariant node descriptors.
- **Steps 1–3:** candidate matching, maximum consistent set and pose with an explicit decline (`lifelong/`, ∼600 lines of Python, unit-tested on synthetic rooms).
- **Evaluation:** the two-visit protocol on the full validation split (47 rooms, 110 pairs), our method and four baselines (SGAligner, SG-PGM, FPFH+RANSAC, FPFH+TEASER++) on identical inputs; metrics, per-pair logs and figures are generated by scripts.

3.1 Scene-graph construction

We load the annotated instance PLY, semantic labels and 3DSSG relations of each 3RScan scan. Floor, wall and ceiling segments become structural nodes; every other segment with at least 50 points becomes an object node. A floor plane $(\mathbf{n}, d)$ is fitted to the floor segment and oriented towards the

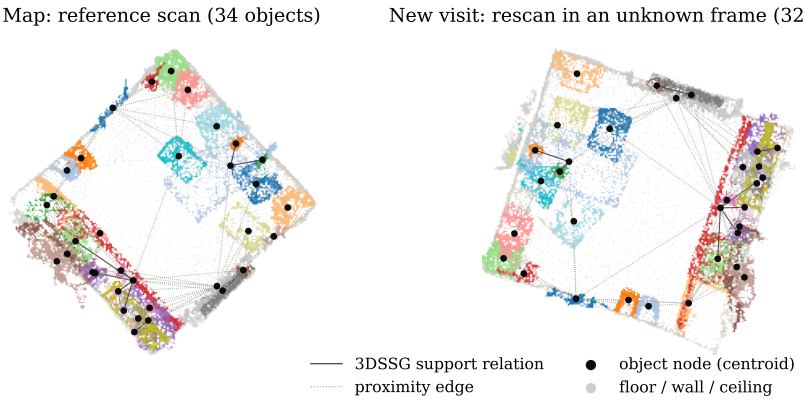


Figure 2: Scene graphs built from the annotated segments of a 3RScan reference scan (left) and one rescan (right), top-down in each scan's own floor frame. Colours mark instances (IDs shared across scans). The rescan is in an unknown frame; 3RScan annotates 11 of its objects as moved and one as removed. Edges: 3DSSG support relations and proximity.

scene. Extents are pose invariant: the vertical extent along $\mathbf{n}$ and the two horizontal extents of the minimum-area bounding rectangle of the footprint (rotating calipers on its convex hull; PCA axes were unstable for square footprints, which a unit test caught), all as 2–98% percentile ranges. Edges are 3DSSG relations plus proximity edges (gap $< 0.3\,\mathrm{m}$). Figure 2 shows the two graphs of one pair.

3.2 Steps 1–3: relocalization

Step 1: candidate matches. For a map object a and a visit object b of the same class,

$$\kappa(a,b) = \sum_{k=1}^{3} \left|\log(e_{a,k}+\epsilon) - \log(e_{b,k}+\epsilon)\right| + \frac{|h_a - h_b|}{\sigma_h}, \qquad \epsilon = 5\,\mathrm{cm},\ \sigma_h = 0.3\,\mathrm{m}, \tag{1}$$

and $\kappa = \infty$ across classes. Each map object keeps its $k = 3$ cheapest visit objects and vice versa (if $\kappa \le 2.5$). Nothing depends on the robot's pose; identical objects receive several candidates and Step 2 picks among them.

Step 2: largest consistent set. Two candidates $m = (a,b)$ and $m' = (a',b')$ are compatible if they are one-to-one ($a \neq a'$, $b \neq b'$) and

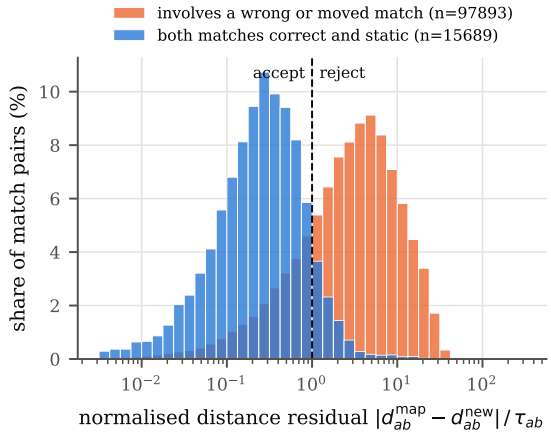
$$\left|\|\mathbf{c}_a - \mathbf{c}_{a'}\| - \|\mathbf{c}_b - \mathbf{c}_{b'}\|\right| \le \tau_{mm'}, \quad \left|(h_a - h_{a'}) - (h_b - h_{b'})\right| \le \tau_{mm'}, \quad \tau_{mm'} = \varepsilon + \beta\,(r_m + r_{m'}), \tag{2}$$

with $\varepsilon = 0.1\,\mathrm{m}$ for sensor and annotation noise and $\beta = 0.1$ scaling with the object radii r, because the centroid of a partly observed large object shifts. The height test rejects sets that only fit with the scene turned upside down; it cannot reject a reflection through a vertical plane, which keeps all distances and heights. The static set S is the maximum clique of the compatibility graph (igraph; ties broken by summed κ). Removing the objects of S and solving again yields further cliques of size ≥ 2: candidate groups of objects that moved together, which Step 4 will verify. On all candidate pairs of the 110 validation pairs (Fig. 3a) the distance test keeps 90.9% of pairs of correct, unchanged matches and 23.3% of the rest; the clique removes most of the remainder, since a wrong match must agree with *every* member of S.

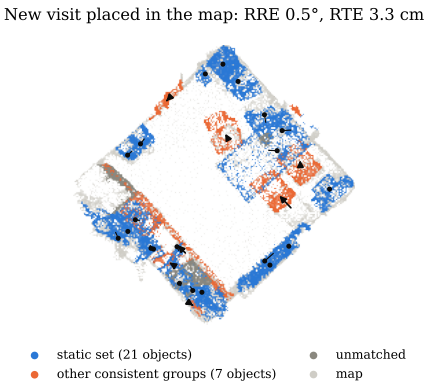
Step 3: pose. With $|S| \ge 3$ and non-collinear centroids (second singular value of the centred centroids $\ge 0.2\,\mathrm{m}\cdot\sqrt{|S|}$) we solve

$$\hat{\mathbf{R}}, \hat{\mathbf{t}} = \arg\min_{\mathbf{R},\mathbf{t}} \sum_{(a,b)\in S} w_{ab}\|\mathbf{R}\mathbf{c}_b + \mathbf{t} - \mathbf{c}_a\|^2 + \lambda\|\mathbf{R}\,\mathbf{n}^N - \mathbf{n}^M\|^2, \qquad w_{ab} = \big(0.1 + \max(r_a, r_b)\big)^{-1}, \tag{3}$$

in closed form (weighted Kabsch/Umeyama [27] with the floor normals as an extra direction pair; with the weights normalized, λ is the weighted variance of the visit centroids, so the normal weighs



(a) Pairwise test on all candidate pairs.



(b) Result for the pair of Fig. 2.

Figure 3: (a) Distance residual over the tolerance $\tau_{mm'}$ for candidate pairs whose two matches are both correct and unchanged (blue) and for all other pairs (orange), 110 validation pairs, distance test only. (b) The visit placed in the map with the estimated pose (RTE 3.3 cm): the static clique (blue, lines to the map centroids) gives the pose. The remaining consistent groups (orange, arrows) are only candidates for Step 4: 4 of their 7 objects are annotated as moved; the others are static objects missed by S, which Step 4 must re-test against $\hat{\mathbf{T}}$.

as much as their spread), then refine with point-to-plane ICP [6] on the points of S and the floor, walls and ceiling (5 cm voxels, correspondence radius $0.3 \to 0.1 \to 0.05$ m). If $|S| < 3$, the centroids are collinear, or the ICP fitness is below 0.3, we return $\perp$. On the development rooms, accepting collinear sets (solvable with the floor normal) only added wrong poses, because those sets were objects that moved together, so the test stays. All thresholds were chosen on 6 *training* rooms (Table 4).

3.3 Two-visit evaluation

Data and protocol. We use all 47 validation rooms of 3RScan [28] with 3DSSG [29], giving 110 pairs of a reference scan and one of its rescans (6–63 objects per map, 3–61 per visit; 383 moved, 96 removed and 54 deformed instances in total). Parameters are unchanged from the training rooms. Each rescan is moved by a stored random transform (uniform rotation, translation $\mathcal{U}(-2, 2)^3$ m) and every method receives exactly these points; $\mathbf{T}^\star$ comes from the 3RScan alignment and is used only for scoring. A second run gives the rescan already aligned to the map, which is the setting in which the learned baselines were trained and evaluated.

Baselines. We chose the strongest methods with runnable 3RScan code: SG-PGM [31] and SGAligner [22] with their released 3RScan-trained checkpoints (our rooms are from the validation split, so there is no train/test leakage), and the training-free FPFH+RANSAC [21, 7] and FPFH+TEASER++ [32], both followed by ICP. SG-Reg [13] (only a ScanNet model and a C++ backend are released) and SGAligner++ [26] (no code) are discussed from their papers. Two fixes were needed for a fair run (details in the appendix): SGAligner's GC-RANSAC needs score-sorted correspondences for its PROSAC sampler (unsorted input gave no model in 53% of aligned pairs), and FPFH normals are oriented towards the scan centre (Open3D's default sign depends on the frame and halved TEASER++'s success under rotation). Both fixes were diagnosed on validation pairs; unpatched numbers are not reported.

Results (Table 2 and Fig. 4). (i) *Invariance.* At 20/20 and in RR our scores are identical with and without the random transform. At 10/10 they drop from 78.2% to 72.7%: translation error is measured at the frame origin, which the transform moves metres away from the scene, so a sub-degree rotation error adds centimetres. The learned aligners collapse under the transform (SGAligner 79%→32%, SG-PGM 75%→34%): their node encoders consume object points in world coordinates, and SG-PGM's published random-transform evaluation rotated only the registration points. (ii) *Accuracy and cost.* Ours lies within the 95% intervals of both FPFH methods (Wilson: ours 82–94%, RANSAC 87–97%);

Table 2: Relocalization on the 110 validation pairs (random transform). Ret. = pose returned; Wrong = returned but outside 20 cm/20°; RR = RMSE < 0.2 m; median errors over returned poses; pose time per pair (our graph construction adds 0.11 s); Aligned = 20/20 with the rescan pre-aligned.

Method	Ret. %	Wrong %	10/10	20/20	RR	RRE (°)	RTE (cm)	Time (s)	Aligned
SGAligner [22]	40.9	9.1	27.3	31.8	32.7	1.45	5.4	3.37	79.1
SG-PGM [31]	100	66.4	27.3	33.6	35.5	17.9	68.3	4.09	74.5
FPFH + TEASER++ [32]	100	9.1	75.5	90.9	93.6	0.98	3.8	1.82	90.9
FPFH + RANSAC [21]	100	6.4	**78.2**	**93.6**	**96.4**	**0.96**	**3.7**	4.83	92.7
Ours, centroid pose only	96.4	11.8	59.1	84.5	89.1	1.63	7.2	<0.01	83.6
Ours (Steps 1–3)	93.6	**4.5**	72.7	89.1	91.8	0.97	4.0	**0.13**	89.1

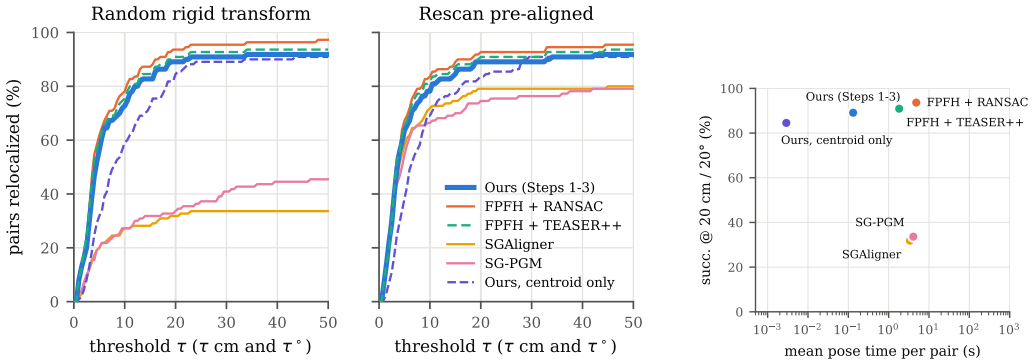


Figure 4: Left: pairs relocalized as the threshold grows (τ cm and $\tau°$ jointly), 110 validation pairs, random transform and pre-aligned. Right: success at 20 cm/20° against mean pose time.

RANSAC succeeds on 5 pairs we fail and we on none it fails (McNemar $p = 0.06$). Ours returns the fewest wrong poses (5, against 7–73 for the baselines): 7 of our 12 failures are declines. Steps 1–2 take 2.6 ms on 3–120 candidates, ICP 0.13 s and graph construction 0.11 s: about 20× less than RANSAC, on annotated segments. ICP lowers the centroid-only error (7.2→4.0 cm) and the wrong poses (13→5). Static-set precision/recall against unchanged instances is 72.6/65.6%; many "errors" are objects annotated as moved that shifted only centimetres. (iii) *Failures* (Table 3, appendix). One café room has a row of identical table–chair–menu sets and no walls: one rescan is matched one set off and another is flipped by 180° about the vertical; both pass the fitness gate, which the floor dominates. Three near-misses have RRE ≈ 4°, RTE 22–34 cm. FPFH+RANSAC solves 5 of our 7 declines from walls and floor, which motivates a structure-only fallback.

4 Timeline for the Remainder of the Project

Dates	Work package	Deliverable
Oct 27–Nov 2	**Done:** full 3RScan validation split (47 rooms, 110 pairs) for all methods; RGB-D sequences of all 157 validation scans downloaded.	Table 2
Nov 3–9	Floor/wall fallback for declines; repeated-layout check from walls. **Step 4**: motion of each remaining clique; single-object moves by ICP. **Step 5**: unmatched map objects projected into the depth frames with $\hat{\mathbf{T}}$: seen through ⇒ removed, occluded or out of view ⇒ not observed; added only over observed free space.	Change P/R per type
Nov 10–16	**Step 6**: per-object map edit with change history and free-space record; multi-visit setting (rooms with ≥ 3 scans); map size and share of objects at the correct place per visit.	Multi-visit curves
Nov 17–26	Predicted segments (Mask3D [25] or SceneGraphFusion [30], with many-to-one candidates); ablations; final report (Nov 23); demo (Nov 26).	Report, demo

References

[1] Aikaterini Adam, Torsten Sattler, Konstantinos Karantzalos, and Tomas Pajdla. Objects can move: 3D change detection by geometric transformation consistency. In *European Conference on Computer Vision (ECCV)*, 2022.

[2] Iro Armeni, Zhi-Yang He, JunYoung Gwak, Amir R. Zamir, Martin Fischer, Jitendra Malik, and Silvio Savarese. 3D scene graph: A structure for unified semantics, 3D space, and camera. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.

[3] Daniel Barath and Jiří Matas. Graph-cut RANSAC. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

[4] Paul J. Besl and Neil D. McKay. A method for registration of 3-D shapes. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 14(2):239–256, 1992.

[5] Gang Chen, Sebastián Barbas Laina, Stefan Leutenegger, and Javier Alonso-Mora. OpenSGA: Efficient 3D scene graph alignment in the open world. *arXiv preprint arXiv:2605.10484*, 2026.

[6] Yang Chen and Gérard Medioni. Object modelling by registration of multiple range images. *Image and Vision Computing*, 10(3):145–155, 1992.

[7] Martin A. Fischler and Robert C. Bolles. Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM*, 24(6):381–395, 1981.

[8] Abel Gawel, Carlo Del Don, Roland Siegwart, Juan Nieto, and Cesar Cadena. X-View: Graph-based semantic multi-view localization. *IEEE Robotics and Automation Letters*, 3(3):1687–1694, 2018.

[9] Qiao Gu, Alihusein Kuwajerwala, Sacha Morin, Krishna Murthy Jatavallabhula, Bipasha Sen, Aditya Agarwal, Corban Rivera, William Paul, Kirsty Ellis, Rama Chellappa, Chuang Gan, Celso Miguel de Melo, Joshua B. Tenenbaum, Antonio Torralba, Florian Shkurti, and Liam Paull. ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.

[10] Nathan Hughes, Yun Chang, and Luca Carlone. Hydra: A real-time spatial perception system for 3D scene graph construction and optimization. In *Robotics: Science and Systems (RSS)*, 2022.

[11] Giseop Kim and Ayoung Kim. LT-mapper: A modular framework for LiDAR-based lifelong mapping. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2022.

[12] Ming Liao, Chao Ye, Jianing Fei, and Weiyang Lin. DSG: Dynamic 3D scene graph construction for embodied agents in changing indoor environments. arXiv preprint arXiv:2609.00619, 2026.

[13] Chuhao Liu, Zhijian Qiao, Jieqi Shi, Ke Wang, Peize Liu, and Shaojie Shen. SG-Reg: Generalizable and efficient scene graph registration. *IEEE Transactions on Robotics*, 2025. arXiv:2504.14440.

[14] Samuel Looper, Javier Rodriguez-Puigvert, Roland Siegwart, Cesar Cadena, and Lukas Schmid. 3D VSG: Long-term semantic scene change prediction through 3D variable scene graphs. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.

[15] Parker C. Lusk, Kaveh Fathian, and Jonathan P. How. CLIPPER: A graph-theoretic framework for robust data association. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.

[16] Joshua G. Mangelson, Derrick Dominic, Ryan M. Eustice, and Ram Vasudevan. Pairwise consistent measurement set maximization for robust multi-robot map merging. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2018.

[17] José Neira and Juan D. Tardós. Data association in stochastic mapping using the joint compatibility test. *IEEE Transactions on Robotics and Automation*, 17(6):890–897, 2001.

[18] Mason B. Peterson, Yixuan Jia, Yulun Tian, Annika Thomas, and Jonathan P. How. ROMAN: Open-set object map alignment for robust view-invariant global localization. In *Robotics: Science and Systems (RSS)*, 2025.

[19] Zheng Qin, Hao Yu, Changjian Wang, Yulan Guo, Yuxing Peng, and Kai Xu. Geometric transformer for fast and robust point cloud registration. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[20] Antoni Rosinol, Andrew Violette, Marcus Abate, Nathan Hughes, Yun Chang, Jingnan Shi, Arjun Gupta, and Luca Carlone. Kimera: From SLAM to spatial perception with 3D dynamic scene graphs. *The International Journal of Robotics Research*, 40(12–14):1510–1546, 2021.

[21] Radu Bogdan Rusu, Nico Blodow, and Michael Beetz. Fast point feature histograms (FPFH) for 3D registration. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2009.

[22] Sayan Deb Sarkar, Ondrej Miksik, Marc Pollefeys, Daniel Barath, and Iro Armeni. SGAligner: 3D scene alignment with scene graphs. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

[23] Lukas Schmid, Marcus Abate, Yun Chang, and Luca Carlone. Khronos: A unified approach for spatio-temporal metric-semantic SLAM in dynamic environments. In *Robotics: Science and Systems (RSS)*, 2024.

[24] Lukas Schmid, Jeffrey Delmerico, Johannes Schönberger, Juan Nieto, Marc Pollefeys, Roland Siegwart, and Cesar Cadena. Panoptic multi-TSDFs: A flexible representation for online multi-resolution volumetric mapping and long-term dynamic scene consistency. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2022.

[25] Jonas Schult, Francis Engelmann, Alexander Hermans, Or Litany, Siyu Tang, and Bastian Leibe. Mask3D: Mask transformer for 3D semantic instance segmentation. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.

[26] Binod Singh, Sayan Deb Sarkar, and Iro Armeni. SGAligner++: Cross-modal language-aided 3D scene graph alignment. *arXiv preprint arXiv:2509.20401*, 2025.

[27] Shinji Umeyama. Least-squares estimation of transformation parameters between two point patterns. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 13(4):376–380, 1991.

[28] Johanna Wald, Armen Avetisyan, Nassir Navab, Federico Tombari, and Matthias Nießner. RIO: 3D object instance re-localization in changing indoor environments. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.

[29] Johanna Wald, Helisa Dhamo, Nassir Navab, and Federico Tombari. Learning 3D semantic scene graphs from 3D indoor reconstructions. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

[30] Shun-Cheng Wu, Johanna Wald, Keisuke Tateno, Nassir Navab, and Federico Tombari. SceneGraphFusion: Incremental 3D scene graph prediction from RGB-D sequences. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

[31] Yaxu Xie, Alain Pagani, and Didier Stricker. SG-PGM: Partial graph matching network with semantic geometric fusion for 3D scene graph alignment and its downstream tasks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[32] Heng Yang, Jingnan Shi, and Luca Carlone. TEASER: Fast and certifiable point cloud registration. *IEEE Transactions on Robotics*, 37(2):314–333, 2021.

[33] Xiyu Zhang, Jiaqi Yang, Shikun Zhang, and Yanning Zhang. 3D registration with maximal cliques. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[34] Liyuan Zhu, Shengyu Huang, Konrad Schindler, and Iro Armeni. Living scenes: Multi-object relocalization and reconstruction in changing 3D environments. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

A Appendix: Failure Cases and Implementation Details

Table 3: Our 12 failures on the 110 validation pairs (random transform). RANSAC: whether FPFH+RANSAC succeeds on the same pair.

Ref. / rescan	Outcome	$\lvert S\rvert$	RANSAC
0cac7540 / 0cac7532	wrong: one-set shift	8	–
0cac7540 / 0cac7534	wrong: 179° flip	5	–
137a8158 / 5630cfd3	near-miss: 4.6°, 34 cm	11	–
137a8158 / f2c76fe9	near-miss: 3.5°, 23 cm	8	–
c7895f27 / c7895f29	near-miss: 4.1°, 22 cm	19	–
10b17940 / c2d99343	$\perp$: too few static	2	–
6bde60cb / 6bde60cd	$\perp$: too few static	1	–
8e0f1c28 / 20c993bd	$\perp$: degenerate	3	✓
8e0f1c28 / 20c993bf	$\perp$: degenerate	3	✓
137a8158 / f2c76fed	$\perp$: fitness 0.15	4	✓
6bde6081 / 6bde607b	$\perp$: fitness 0.19	7	✓
6bde60cb / 6bde60cf	$\perp$: fitness 0.29	5	✓

Table 4: Parameters of our method (`configs/default.yaml`), chosen on 6 training rooms (22 pairs). A sweep over ε, β, k and the cost gate gave 19/22 correct and no wrong poses almost everywhere.

Step	Parameter	Value
Graph	min. segment size	50 points
1	candidates per object k	3 (both directions)
1	extent offset ϵ, height scale σ_h	5 cm, 0.3 m
1	cost gate	$\kappa \leq 2.5$, same class
2	tolerance ε, β	0.10 m, 0.10
2	moved-group min. size	2
3	min. static objects / spread	3 / 0.2 m
3	ICP voxel, radii	5 cm; 0.3, 0.1, 0.05 m
3	min. ICP fitness	0.3

Baseline setup. Our adapter exports the pairs to SGAligner's data layout and runs its own preprocessing; SG-PGM reads the same files. One conda environment (PyTorch 2.7.1/CUDA 12.8, RTX 2080 Ti) runs SGAligner ($K = 2$ node matches, 3DMatch GeoTransformer, GC-RANSAC) and SG-PGM (released weights, GeoTransformer backbone). SGAligner's own preprocessing builds the per-scan pickles from the exported full scans (attribute vocabulary 164, relations 41, as in the checkpoints); 3DSSG "inside" relations are dropped because that preprocessing rejects them. SG-PGM's internal augmentation is disabled since the transform is already in the data. Source commits, patches and package versions are in the repository (`baselines/`).

Data and models. Scans come from the Hugging Face mirror `maartenImjalli143/3RScan`; metadata and 3DSSG from the official release. No models were trained, so there are no Hugging Face or WandB model artefacts.

Reproduction. `bash scripts/run_full_val.sh` downloads the split and runs pair generation, all methods and the evaluation; `scripts/make_figures.py -config configs/full_val.yaml -results results/full_val` regenerates the figures.